15B
CHEMICAL EQUILIBRIUM

UNIT 02: HOW REACTIONS HAPPEN

CHEMISTRY 1310

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LEARNING OBJECTIVES

CHAPTER 15

1. Define and explain the condition of equilibrium.
2. Determine equilibrium constant expressions and their values.
3. Use equilibrium constant expressions as formulas.
4. Manipulate equilibrium constant expressions based on changes to the chemical equation.
5. Define and explain the reaction quotient.
6. Predict the direction of a reaction using the relationship between the reaction quotient and the equilibrium constant.
7. Use ICE tables to solve equilibrium problems.
8. Identify the reaction conditions that shift the equilibrium position of a reaction.

The Reaction Quotient

15.4

For a chemical reaction, the **equilibrium constant** (K) is a ratio of (the concentrations of) products to reactants *at equilibrium*.

At equilibrium	$a\mathrm{A} + b\mathrm{B} \rightleftharpoons c\mathrm{C} + d\mathrm{D}$	$\boldsymbol{K} = \dfrac{[\mathrm{C}]_{\mathrm{eq}}^{c}[\mathrm{D}]_{\mathrm{eq}}^{d}}{[\mathrm{A}]_{\mathrm{eq}}^{a}[\mathrm{B}]_{\mathrm{eq}}^{b}}$

If the reaction is *not at equilibrium*, then this ratio of (the concentrations of) products to reactants is called the **reaction quotient** (Q).

Not at equilibrium	$a\mathrm{A} + b\mathrm{B} \rightleftharpoons c\mathrm{C} + d\mathrm{D}$	$\boldsymbol{Q} = \dfrac{[\mathrm{C}]^{c}\,[\mathrm{D}]^{d}}{[\mathrm{A}]^{a}\,[\mathrm{B}]^{b}}$

All equilibrium constants are reaction quotients, but not all reaction quotients are equilibrium constants.

The Reaction Quotient

15.4

The value of ***Q* relative to *K*** is a measure of the progress of a reaction toward equilibrium!

Consider a generic reaction: $2A \rightleftharpoons A_2$

Reaction must proceed *toward products* to reach equilibrium.

"shifts to the right"

$Q < K$

Reaction is at equilibrium.

$Q = K$

Reaction must proceed *toward reactants* to reach equilibrium.

"shifts to the left"

$Q > K$

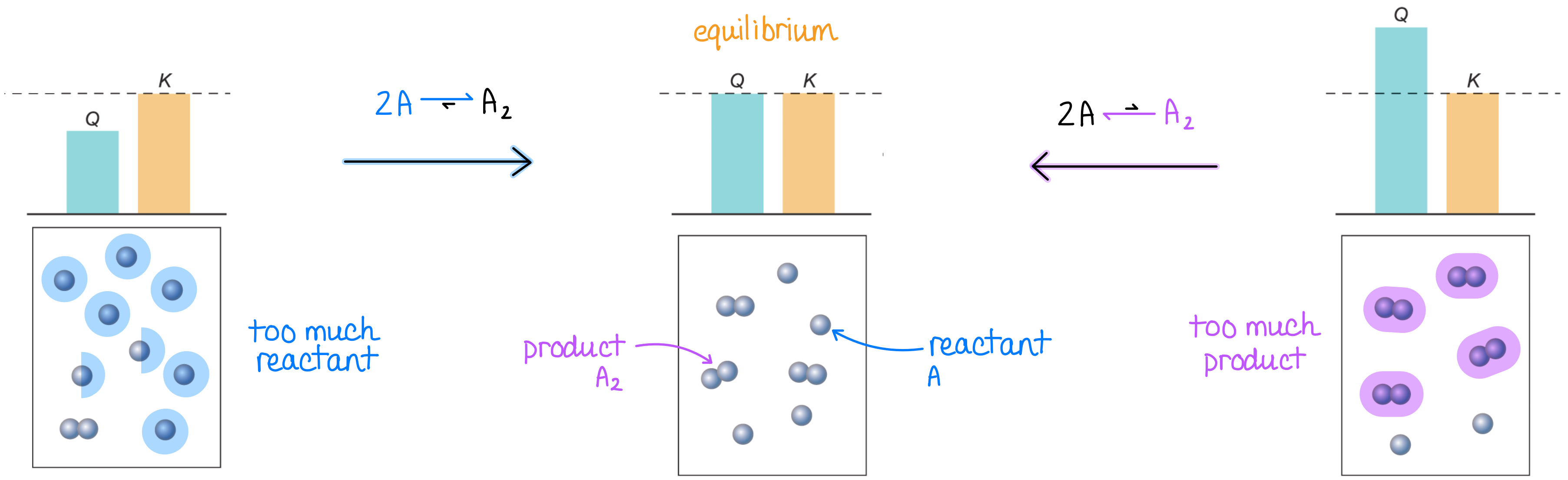


FIGURE 15.5

Calculations Using ICE Tables

15.5

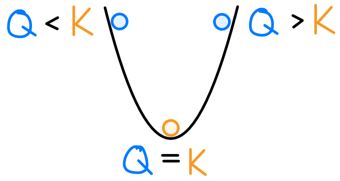
Unlike one-directional (→) chemical reactions, *stoichiometry* involving chemical reactions that come to *equilibrium* ($\rightleftharpoons$) are more complex because Q and K must be accounted for.

To streamline these equilibrium calculations, chemists employ the following basic steps that result in an ICE table (or ICE chart).

1) Write a balanced chemical equation for the equilibrium.
2) Write the corresponding equilibrium constant (K) expression.
3) Construct a *table of concentrations* for relevant species.
4) Use Q vs. K to determine whether reaction shifts *right* ($Q < K$) or *left* ($Q > K$).

shifts right
consume reactant
make product

shifts left
consume product
make reactant



		aA	+	bB	$\rightleftharpoons$	cC	+	dD
Initial	I							
Change	C							
Equilibrium	E							

step 1: write K expression
step 2: set up ICE chart
step 3: use Q vs. K to get shift
step 4: plug "E" row into K
step 5: solve for "x"
step 6: use "x" to solve problem

EXAMPLE PROBLEM

Learning Objective(s): 15.6, 15.7

In a rigid 1.00 L container, 0.500 mol of H_2 (initial) and 0.500 mol of I_2 (initial) are allowed to react at a constant temperature. At equilibrium, the mixture produces 0.900 mol of HI.

Use an ICE chart to determine the value of the equilibrium constant (K) at this temperature.

$$H_2(g) + I_2(g) \rightleftharpoons 2\ HI(g)$$

	$H_2(g)$	$I_2(g)$	$2\ HI(g)$
I	0.500 M	0.500 M	0 M
C	$-x$	$-x$	$+2x$
E	$0.500-x$	$0.500-x$	$2x = 0.900$ M

$x = 0.450$ M

$$K = \frac{[HI]^2}{[H_2][I_2]} = \frac{(2x)^2}{(0.500-x)(0.500-x)} = \frac{(0.900)^2}{(0.500-0.450)(0.500-0.450)} = \frac{(0.900)^2}{(0.050)(0.050)}$$

$K = 324$

How to know which side to subtract/add?

$Q = 0 < K$; shifts right: consume reactants (subtract)
make products (add)

step 1: write K expression
step 2: set up ICE chart
step 3: use Q vs. K to get shift
step 4: plug "E" row into K
step 5: solve for "x"
step 6: use "x" to solve problem

EXAMPLE PROBLEM

Learning Objective(s): 15.6, 15.7

Calculate the equilibrium concentrations (in M) of all species given the following initial conditions for the reaction carried out at a constant temperature where $K_c = 5.8$.

	$CO(g)$ +	$H_2O(g)$ ⇌	$H_2(g)$ +	$CO_2(g)$
I	0.0125 M	0.0125 M	0 M	0 M
C	−x	−x	+x	+x
E	0.0125 −x	0.0125 −x	x	x

How to know which side to subtract/add?

Q = 0 < K ; shifts right : consume reactants (subtract)
make products (add)

at equilibrium

$$K = \frac{[H_2][CO_2]}{[H_2O][CO]}$$

$$5.8 = \frac{(x)(x)}{(0.0125 - x)(0.0125 - x)}$$

$$5.8 = \frac{x^2}{(0.0125 - x)^2}$$

$x = 0.0088_3$ M

now, plug x into E row

at equilibrium

$[H_2] = x = 0.0088$ M

$[CO_2] = x = 0.0088$ M

$[H_2O] = 0.0125 - x = 0.0037$ M

$[CO] = 0.0125 - x = 0.0037$ M

step 1: write K expression
step 2: set up ICE chart
step 3: use Q vs. K to get shift
step 4: plug "E" row into K
step 5: solve for "x"
step 6: use "x" to solve problem

IN-CLASS QUESTION 1

Learning Objective(s): 15.6, 15.7

A mixture of 0.384 M H_2O, 0.384 M Cl_2O, and 0.652 M HClO is placed in a vessel at 25 °C. What is the concentration (in M) of HClO at equilibrium?

$$H_2O(g) + Cl_2O(g) \rightleftharpoons 2\,HClO(g) \qquad K_c = 0.0900$$

	$H_2O(g)$	$Cl_2O(g)$	$2\,HClO(g)$
I	0.384 M	0.384 M	0.652 M
C	+x	+x	−2x
E	0.384 M + x	0.384 M + x	0.652 −2x

How to know which side to subtract/add?

$$Q = \frac{(0.652)^2}{(0.384)(0.384)}$$

$Q = 2.88$

$Q > K$; shifts left : consume products (subtract)
make reactants (add)

at equilibrium

$$K = \frac{[HClO]^2}{[H_2O][Cl_2O]}$$

$$0.0900 = \frac{(0.652 - 2x)^2}{(0.384 + x)(0.384 + x)}$$

$$0.0900 = \frac{(0.652 - 2x)^2}{(0.384 + x)^2}$$; take square root both sides

$$0.0300 = \frac{0.652 - 2x}{0.384 + x}$$

$x = 0.233_4$ M

now, plug x into E row

$[HClO] = 0.652 - 2x$

$= 0.652 - 2(0.233_4)$

$[HClO] = 0.185$ M

IN-CLASS QUESTION 2

step 1: write K expression
step 2: set up ICE chart
step 3: use Q vs. K to get shift
step 4: plug "E" row into K
step 5: solve for "x"
step 6: use "x" to solve problem

Learning Objective(s): 15.6, 15.7

"p" for partial pressures: $P_i = \frac{n_i RT}{V}$

The following reaction has $K_p = 0.263$ at 1000 K. What is the total pressure (in atm) in a 10.0 L flask when 25.0 g of C(*s*) and 4.50 g of H_2(*g*) are placed inside and allowed to equilibrate at 1000 K?

recall
solids and liquids not in K!

	C(*s*) +	2 H_2 (*g*) ⇌	CH_4(*g*)
I	n/a	18.3_2 atm	0 atm
C	n/a	$-2x$	$+x$
E	n/a	$18.3_2 - 2x$	x

$K_p = 0.263$

$K_p = \frac{P_{CH_4}}{(P_{H_2})^2}$; at equilibrium

$0.263 = \frac{x}{(18.3_2 - 2x)^2}$

$0 = 1.05_2 x^2 - 20.2_6 x + 88.2_3$; solve quadratic

$x = 12.6_2$ atm × unphysical solution

and

$x = 6.64_5$ atm ✓ physical solution

get partial pressure

$P_{H_2} = \frac{n_{H_2} RT}{V}$

$= \frac{\left(4.50\text{ g } H_2 \times \frac{1\text{ mol } H_2}{2.016\text{ g } H_2}\right)\left(0.08206 \frac{\text{L}\cdot\text{atm}}{\text{mol}\cdot\text{K}}\right)(1000\text{K})}{10.0\text{ L}}$

$P_{H_2} = 18.3_2$ atm

plug "x" into "E" row

$P_{CH_4} = x$

$P_{CH_4} = 6.64_5$ atm

$P_{H_2} = 18.3_2 - 2x$

$P_{H_2} = 18.3_2\text{ atm} - 2(6.64_5\text{ atm}) = 5.0_2\text{ atm}$

Get total pressure

$P_{tot} = P_{H_2} + P_{CH_4}$

$= 5.0_2\text{ atm} + 6.64_5\text{ atm}$

$P_{tot} = 11.7$ atm

How to know which side to subtract/add?

$Q = 0 < K$; shifts right : consume reactants (subtract)
make products (add)

BONUS QUESTION

Assessed on: Summer 2023, Final Exam

$$N_2H_4(g) + H_2(g) \rightleftharpoons 2\ NH_3(g)$$

What is the equilibrium concentration of NH_3 if the initial concentration of N_2H_4 is 0.550 M and initial concentration of H_2 is 0.850 M. At this particular temperature, $K = 0.520$ for the reaction.